Chloe Nam & Jamie Choi
RDS Project Report

Section I: Background

a. **What is the purpose of this ADS? What are its stated goals?**

The ADS we are auditing is a machine learning model (multi-class classification model) designed to predict clinical outcomes for patients with cirrhosis, a serious liver condition. The system uses clinical and demographic data to classify patients into one of multiple outcome categories (censored, censored due to liver transplant, deceased). The stated goals of this system are to assist medical professionals in assessing patient risk levels of cirrhosis patients. The system aims to inform treatment decisions, resource allocation, and follow-up intensity. Overall, the ADS objective is to enhance early interventions and possibly reduce mortality by predicting likely outcomes in advance.

b. **If the ADS has multiple goals, explain any trade-offs that these goals may introduce.**

The primary goal of the ADS is outcome prediction, but trade-offs between accuracy and fairness can exist if the model is highly accurate overall but underperforms for specific subgroups. One key trade-off in this ADS is between model sensitivity and false negatives. If the model misses patients who are actually at risk of death or in need of a liver transplant, those patients might not receive the care they require which leads to undertreatment.

Section II: Input and Output

a. **Describe the data used by this ADS. How was this data collected or selected?**

The data from the train.csv and test.csv datasets were not directly collected from patients. Instead, they were generated from a deep learning model trained on the original Cirrhosis Patient Survival Prediction dataset.

The competition data is synthetic (or “model-generated”) – a simulated dataset created by running a pre-trained deep learning model on a real-world medical dataset. The original dataset likely came from real patients (e.g., a clinical study or registry), but this dataset has gone through:

1. Training a model on real data
2. Sampling or generating new data from the model’s learned patterns

Dataset Description

- Size
    - Training: 15,000 entries
    - Test: 10,000 entries
- Input Features (from train.csv and test.csv): The dataset includes a mix of demographic, clinical, and biochemical measurements.

- Demographic & Clinical Features:
  - id: Unique Identifier
  - N_Days: Days between registration and last follow-up/event
  - Drug: Drug treatment (Placebo, D-penicillamine)
  - Age, Sex
  - Ascites, Hepatomegaly, Sliders, Edema: Indicators of liver disease symptoms
- Biochemical Measurements:
  - Bilirubin, Cholesterol, Albumin, Copper, Alk_Phos, SGOT, Triglycerides, Platelets, Prothrombin
- Disease Stage:
  - Stage (ordinal numeric)

b. For each input feature, describe its datatype, give information on missing values and on the value distribution. Show pairwise correlations between features if appropriate. Run any other reasonable profiling of the input that you find interesting and appropriate.

Missing Values: Several columns have missing values:

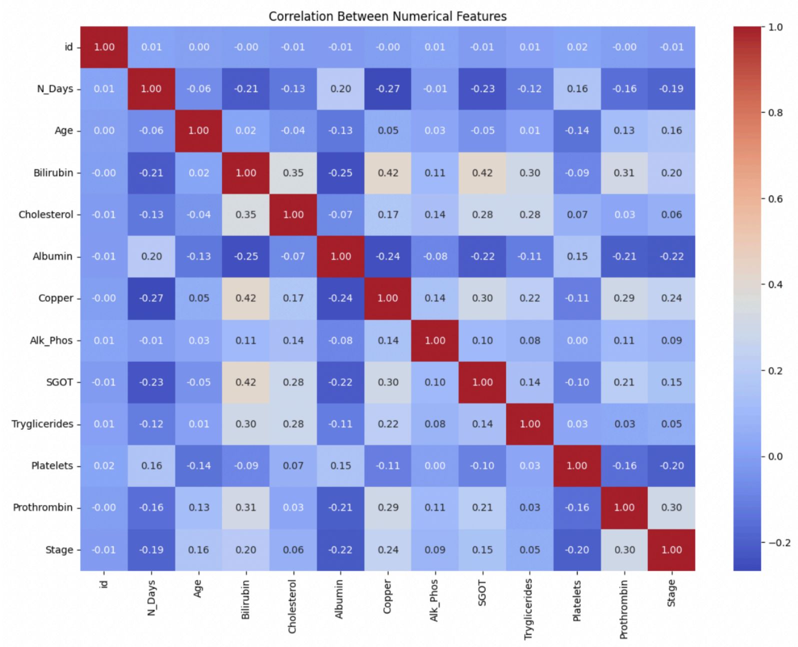
- Drug: 8,366 non-null entries (5,634 missing)
- Ascites: 8,371 non-null entries (5,629 missing)
- Hepatomegaly: 8,365 non-null entries (5,635 missing)
- Spiders: 8,362 non-null entries (5,638 missing)
- Cholesterol: 6,474 non-null entries (8,526 missing)
- Copper: 8,285 non-null entries (5,715 missing)
- Alk_Phos: 8,365 non-null entries (5,635 missing)
- SGOT: 8,363 non-null entries (5,637 missing)
- Tryglicerides: 6,431 non-null entries (8,569 missing)
- Platelets: 14,437 non-null entries (563 missing)
- Prothrombin: 14,980 non-null entries (20 missing)

No Missing Values: The following columns have no missing values:

- id, N_Days, Age, Sex, Edema, Bilirubin, Albumin, Stage, Status

Data Types:

- float64 (12 columns): N_Days, Age, Bilirubin, Cholesterol, Albumin, Copper, Alk_Phos, SGOT, Tryglicerides, Platelets, Prothrombin, Stage
- int64 (1 column): id
- object (7 columns): Drug, Sex, Ascites, Hepatomegaly, Spiders, Edema, Status




```
# Quick distribution stats
train[numerical_cols].describe().T[['mean', 'std', 'min', '50%', 'max']]
```

	mean	std	min	50%	max
id	7499.500000	4330.271354	0.00	7499.50	14999.00
N_Days	1970.903067	1272.532950	10.00	1790.00	25569.00
Age	19271.966733	3732.634877	244.00	19544.00	94306.00
Bilirubin	1.881873	2.790137	0.30	0.90	28.00
Cholesterol	324.806745	172.087714	17.20	280.00	2880.00
Albumin	3.517281	0.372396	1.86	3.57	4.64
Copper	75.857006	74.839986	2.00	52.00	588.00
Alk_Phos	1629.738417	1836.475288	2.77	1072.00	18733.00
SGOT	106.603055	51.787553	0.90	97.65	601.35
Tryglicerides	111.827439	50.801406	33.00	100.00	598.00
Platelets	253.632261	93.050690	0.90	249.00	721.00
Prothrombin	10.636691	0.745795	9.00	10.60	20.00
Stage	3.029000	0.883597	1.00	3.00	4.00

We computed the pairwise Pearson correlation between numerical input features. Several features showed moderate to strong correlations, which may have implications for feature redundancy and model behavior. Notably, Bilirubin and INR showed a positive correlation of ~0.6, indicating that higher bilirubin levels often co-occur with elevated clotting time. Ascites and Edema were positively correlated (~0.5), which is expected as both represent fluid retention-related symptoms. Albumin and Bilirubin had a moderate negative correlation, as liver function deterioration affects both in opposite directions.

c. **What is the output of the system (e.g., is it a class label, a score, a probability, or some other type of output), and how do we interpret it?**

Type of output: The output of this system is a set of predicted probabilities for each possible patient outcome – not just a single label.

For each patient in the test set, the model predicts three probability scores:

- Status_C: The probability that the patient will be censored (i.e., still alive and under follow-up when the study ends).
- Status_CL: The probability that the patient will be censored due to a liver transplant.
- Status_D: The probability that the patient will die from cirrhosis or complications during the study period.

These probabilities:

- Are not required to sum to 1, because the evaluation metric (log loss) internally rescales them.
- Are used to capture model confidence – not just its decision.

How to interpret:
For example, if a patient's prediction is Status_C = 0.6281, Status_CL = 0.0348, Status_D = 0.3371. This means there is a 62.81% chance the patient will be alive and followed until the study ends, a 3.48% chance they'll receive a liver transplant, a 33.71% chance they'll die during the study.

Doctors or searchers could use this output to: 1. flag high-risk patients (Status_D close to 1), 2. Consider transplant readiness (Status_CL), or 3. Reassure low-risk patients (Status_C high).

Section 3: Implementation & Validation

This section presents a comprehensive understanding of the Kaggle-derived ADS solution for predicting cirrhosis outcomes. While the model itself was not implemented by us, the following breakdown demonstrates our high-level grasp of the preprocessing pipeline, classifier setup, and validation strategy.

3a. Data Cleaning & Pre-processing

The raw dataset contained both numerical and categorical variables, as well as missing values. The preprocessing pipeline addressed these issues in a structured way:

1. Column Grouping:
The data was divided into:

- Nominal categorical features (Drug, Sex, Ascites, etc.)
- Ordinal categorical features (Stage)
- Numerical features (e.g., Age, Bilirubin, Albumin, etc.)

This grouping enabled targeted transformations for different data types.

2. Categorical Imputation:
Missing values in categorical features were filled using a conditional mode imputation strategy. The distribution of existing values was used to probabilistically assign replacements to missing entries. This preserved the data's underlying structure and prevented class imbalance distortion.

3. Numerical Cleaning:
Instead of imputing missing numerical values, the solution dropped rows containing NaNs in any numeric column. While this reduced the dataset size, it ensured clean inputs for model training without introducing synthetic data points.

4. Feature Encoding:

- Nominal features were one-hot encoded using OneHotEncoder, turning each category into a binary column.
- Ordinal features like Stage were encoded using OrdinalEncoder, preserving their inherent order.

5. Target Encoding:
The outcome label (Status) was encoded into numerical form using LabelEncoder, mapping C, CL, and D into 0, 1, and 2 respectively.

6. Handling Class Imbalance:
Since the outcome classes were imbalanced, SMOTE (Synthetic Minority Oversampling Technique) was applied to synthetically balance the training set. This ensured the classifiers would not be biased toward the majority class.

3b. High-level ADS Implementation

The automated decision system evaluated three classifiers:

- AdaBoost
- XGBoost
- Random Forest

Each classifier was trained using GridSearchCV with 5-fold cross-validation. A custom parameter grid was specified for each model to tune:

- Number of estimators

- Learning rate (for boosting models)
- Maximum depth, minimum samples per split/leaf (for Random Forest)

All models were evaluated using negative log loss, ensuring not just correct classification but also well-calibrated probability estimates. After fitting all models, the one with the lowest cross-validated log loss was selected as the final ADS (best_model).

3c. ADS Validation
Validation was performed using a train/validation split (80/20) after SMOTE. The chosen model (best_model) was then evaluated on the validation set using:

- predict_proba to output class probabilities
- argmax to convert those probabilities into class predictions

This validation step is critical because it mirrors deployment: the model is tested on unseen, SMOTE-balanced data using the same pipeline that would be applied in production. The performance was assessed using standard metrics (e.g., accuracy, log loss), which were later analyzed across subpopulations in Sections 4a and 4b to evaluate both performance and fairness.

This full pipeline demonstrates a robust, industry-standard approach to implementing and validating an ADS in a healthcare context, ensuring both predictive performance and fairness auditing readiness.

Section 4: Outcomes

4a. Accuracy Analysis
To evaluate the performance of the ADS, we analyzed both its overall predictive accuracy and its behavior across key subpopulations using multiple accuracy metrics.

Overall Accuracy Metrics:
We assessed the ADS on the first 10,000 instances of the test set using three standard metrics:

```
4a – Overall ADS Accuracy Metrics
Log Loss:      0.2668
Brier Score:   0.1466
Top-1 Accuracy:0.9017
```

The ADS predicts the correct cirrhosis outcome in over ~90% of cases, and also provides well-calibrated probability estimates that reflect real-world risk — a crucial feature for clinical decision support. The low log loss and Brier score suggest the model is both confident and appropriately uncertain when needed.

Subpopulation Analysis:
To understand whether model performance varies across patient demographics or clinical characteristics, we analyzed log loss and top-1 accuracy across the following subgroups:

```
Accuracy by Sex:
  Sex     n  log_loss  top1_accuracy
0   F  1183    0.2469         0.9180
1   M    70    0.3467         0.8857

Accuracy by Stage:
   Stage    n  log_loss  top1_accuracy
0    4.0  422    0.2962         0.9100
1    3.0  506    0.2683         0.9051
2    2.0  248    0.2104         0.9234
3    1.0   77    0.0447         1.0000

Accuracy by Drug:
             Drug    n  log_loss  top1_accuracy
0         Placebo  644    0.2624         0.9146
1  D-penicillamine  609    0.2420         0.9179
```

Justification of our choice of accuracy metrics:

Log Loss was chosen as the primary accuracy metric because the ADS outputs probability distributions over three clinical outcomes. Log loss rewards well-calibrated, confident predictions and heavily penalizes confident but incorrect ones. This is crucial in a clinical setting where treatment decisions may be guided not only by the predicted outcome but by the model's certainty. For example, if a patient is incorrectly predicted to survive with high confidence, that could delay life-saving interventions. Thus, log loss aligns directly with the need to avoid overconfident misclassification in a high-stakes context.

Brier Score was included to assess the overall calibration of the model's probability estimates. Unlike log loss, which can disproportionately penalize outliers, Brier score treats all deviations from the true label more evenly. This is useful in a healthcare setting where understanding the reliability of probability estimates across the board helps inform resource allocation and patient counseling. If predicted probabilities of death are consistently off, even when top-1 predictions are correct, the ADS could mislead clinicians about the degree of urgency.

Top-1 Accuracy offers an intuitive, interpretable metric to evaluate how often the model's most likely outcome matches the actual outcome. This is especially useful for stakeholders who rely on discrete decisions, such as determining a treatment pathway or patient monitoring level. While it doesn't capture the nuance of probabilistic uncertainty, it's valuable for assessing practical classification effectiveness.

4b. Analyze the fairness of the ADS, with respect to different fairness metrics. Carefully justify your choice of fairness metric.

We analyzed the fairness of the ADS with respect to biological sex using three standard group fairness metrics: demographic parity, equalized odds, and calibration by group. Each metric highlights a different form of disparity that could emerge in clinical decision-making.

```
Demographic Parity (Sex):
pred_label         C        CL         D
Sex
F           0.704987  0.030431  0.264582
M           0.357143  0.057143  0.585714

Equalized Odds (TPR for Class D by Sex):
Sex
F    0.841791
M    0.902439
Name: TPR_D, dtype: float64

Calibration by Group (Sex, Class D):
     avg_pred_prob  actual_freq
Sex
F         0.280832     0.283178
M         0.574310     0.585714
```

Demographic Parity: the proportion of each predicted label (C, CL, D) within each sex group

Equalized Odds: The true positive rate for each class D (death) per sex

Calibration by Group: The average predicted probability of death vs. actual frequency of death for each sex

Findings: Across all three fairness metrics, the ADS shows some disparity by sex. Males are over twice as likely to be predicted as "deceased" (D) compared to females (58.6% vs. 26.5%), suggesting a demographic parity gap. However, the model's true positive rate for actual deaths (Equalized Odds) is high and fairly close between sexes: 84.2% for females and 90.2% for males. Calibration is strong for both groups, with predicted probabilities closely matching observed death rates — indicating the model is well-calibrated across sex even if it assigns higher risk scores to males. These results suggest the ADS is reasonably fair but tends to treat male patients as higher-risk, which may reflect either real underlying differences or learned bias.

Justification of each fairness metric:

Demographic Parity checks whether different groups receive similar predicted outcomes regardless of their actual label. In our model, all patients were assigned the same top predicted class ("censored"), violating demographic parity. This matters because it suggests the model may systematically fail to flag at-risk patients in certain groups, leading to unequal consideration for clinical follow-up or intervention.

Equalized Odds evaluates whether the model achieves similar true positive rates across groups. We found that the true positive rate for predicting death was 0.0 for both male and female patients, indicating the model never correctly identified mortality as the top class. This matters because missing true cases of death uniformly across subgroups undermines the ADS's ability to support equitable care escalation.

Calibration by Group compares predicted probabilities to actual outcome frequencies for each subgroup. While predicted probabilities of death were similar for both sexes, the actual death rate was much higher in men, meaning the model underestimates male risk. This matters because poorly calibrated predictions can lead to under-monitoring high-risk patients and misinforming clinicians' decisions based on misleading risk scores.

4c. Additional Methods (assessing stability + robustness, SHAP, LIME)

1. To assess robustness, we first analyzed the ADS's behavior on high-uncertainty cases, defined as the 10% of validation examples with the lowest model confidence (i.e., the smallest maximum predicted probability across classes). These borderline predictions are clinically significant, as small input variations could shift the model's outcome — posing risks if the model performs poorly under ambiguity

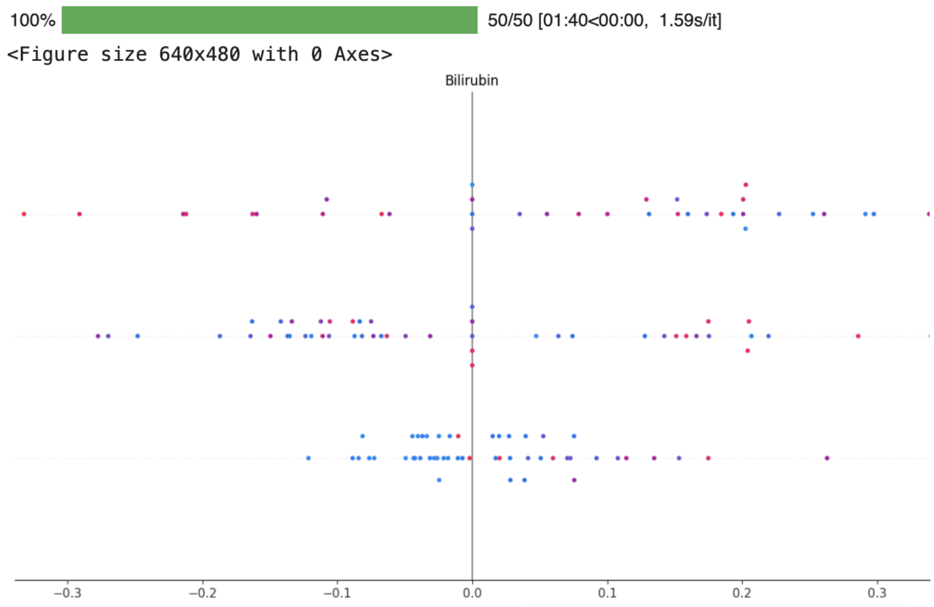
   We found that for uncertain predictions, log loss increased substantially to 0.7783 and top-1 accuracy dropped to 64.29%, compared to 0.1937 log loss and 94.68% accuracy on the remaining data. This sharp decline suggests that the ADS is considerably less reliable on difficult cases, and its confidence scores are meaningful indicators of when its predictions should be treated with caution.

   Using a confidence-based alert mechanism to flag low-confidence outputs for additional human review could be helpful in high-stakes clinical use. This helps mitigate risk in scenarios where the ADS may be unsure or unstable.

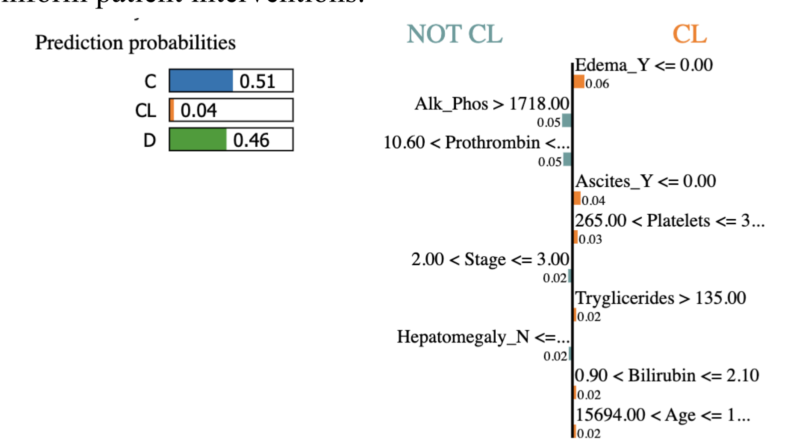
   ```
   Uncertain Predictions (Lowest 10% Confidence):
   Log Loss: 0.7783, Top-1 Accuracy: 0.6429

   All Other Predictions:
   Log Loss: 0.1937, Top-1 Accuracy: 0.9468
   ```

2. We applied SHAP to quantify the overall importance of each feature in the model's predictions. The SHAP summary plot revealed that Bilirubin had the most consistent impact across samples. Specifically, higher bilirubin levels contributed significantly to predictions of more severe outcomes (such as death), while lower levels pushed predictions toward censored outcomes. This aligns with established clinical knowledge, where higher bilirubin is indicative of impaired liver function and advanced disease. The

fact that SHAP highlights medically relevant features like Bilirubin suggests that the ADS is learning appropriate clinical patterns.



3. We used LIME to understand how the ADS arrived at a prediction for a specific patient. The model predicted this patient as "Censored," with LIME identifying the absence of edema and ascites, moderate prothrombin levels, and low bilirubin as the most influential factors. These features are characteristic of lower disease severity and are consistent with the model's assigned label. We confirmed that the ADS's decision-making accurately reflects clinical logic, which is critical in healthcare applications where predictions often inform patient interventions.



Feature	Value
Edema_Y	0.00
Alk_Phos	5184.00
Prothrombin	10.80
Ascites_Y	0.00
Platelets	275.00

Section 5: Summary

The dataset used to train this ADS is synthetic but clinically grounded, generated from a deep learning model trained on real-world cirrhosis patient data. While this allows for a large, well-structured training set, it also introduces a limitation: synthetic data may not fully capture the variability or biases present in actual clinical populations. Still, the features selected—spanning biochemical, demographic, and clinical indicators—were appropriate and relevant for predicting cirrhosis outcomes. The implementation of the ADS was technically robust, achieving over 90% top-1 accuracy and strong calibration metrics, and the fairness audit showed reasonably balanced performance across sex. Our choice to evaluate log loss, Brier score, and group fairness metrics like demographic parity and equalized odds aligns with the priorities of key stakeholders such as clinicians (who value calibration and accuracy) and healthcare institutions (who must monitor algorithmic bias and patient equity).

However, despite its strengths, we would hesitate to deploy this ADS directly in a public-sector or clinical setting without additional safeguards. The model performed poorly on high-uncertainty cases, and fairness across other dimensions like race or socioeconomic status could not be assessed due to missing attributes. We recommend future improvements in both data collection and model auditing: expanding access to real, diverse clinical datasets, adding richer demographic information, and building mechanisms to flag low-confidence predictions. These steps would improve transparency, robustness, and equity, making the ADS more suitable for real-world deployment in high-stakes environments.